

Project Title : Multi-Tenant Course Selling App

Course Name & Code : DBMS - (UCS310)

Degree & Year : B. Tech (2nd Year)

Department / Institute Name : COE / TIET

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1. Introduction

The proposed project is a multi-tenant SaaS-based course selling platform that enables independent educators to create and manage their own branded teaching environments while allowing students to enroll in instructor-specific courses. The domain of online education was selected due to its heavy reliance on structured data, secure user management, and reliable transactional workflows.

A Database Management System plays a crucial role in this application as it involves multiple interrelated entities such as instructors, students, courses, enrollments, folders, and learning content. Compared to a file-based system, a DBMS provides better data consistency, reduced redundancy, efficient querying, and safe concurrent access, which are essential for scalable SaaS platforms.

2. Problem Statement

Independent educators often rely on fragmented or manual systems to manage students, courses, learning content, and paid enrollments. Such approaches lead to data duplication, weak access control, lack of traceability, and poor scalability. The problem addressed in this project is the absence of an integrated, database-driven backend system that can efficiently manage instructor onboarding, student authentication, course lifecycle management, content organization, and enrollment records. A relational DBMS-based solution is required to maintain structured data relationships, enforce constraints, and ensure consistency across all operations.

3. Objectives of the Project

- To design a relational database using E–R data modeling techniques
- To convert the ER model into a well-defined relational schema
- To apply normalization up to Third Normal Form (3NF)
- To implement the database using SQL with proper constraints
- To ensure data integrity using primary keys, foreign keys, and unique constraints
- To support backend CRUD and relational data retrieval operations

4. Scope of the Project

- Instructor registration and authentication
- Student registration under instructor-specific contexts
- Course creation, modification, and listing
- Course content organization using folders and content entities
- Enrollment management linked with payment confirmation

The system involves three primary user roles: **Instructor**, **Student**, and **Admin**. The project focuses mainly on backend database design and implementation, with APIs consuming and managing structured data.

5. Proposed System Description

The proposed system is a SaaS-based backend-driven application where instructors register on the main platform and receive a unique identifier. Students interact within the context of an instructor's domain. The backend manages authentication, authorization, course data, enrollments, and content metadata.

The relational database ensures strong data integrity through enforced constraints, improves efficiency through optimized queries, and enhances consistency by maintaining well-defined relationships among users, courses, and enrollments.

6. Database Design

6.1 Entity–Relationship (ER) Diagram:

The core entities identified in the system are:

- Instructor
- Student
- Admin
- Course
- Enrollment
- CourseFolder
- CourseContent

Key relationships include:

- One Instructor can have multiple Students and Courses
- A Student can enroll in multiple Courses through Enrollment
- A Course can have multiple CourseFolders and CourseContents

6.2 Relational Schema

- **Instructor** (id PK, email UNIQUE, organization UNIQUE, slug UNIQUE)
- **Student** (id PK, instructorId FK, UNIQUE(email, instructorId))
- **Admin** (id PK, email UNIQUE)
- **Course** (id PK, instructorId FK)
- **Enrollment** (id PK, studentId FK, courseId FK, UNIQUE(studentId, courseId))
- **CourseFolder** (id PK, courseId FK, UNIQUE(courseId, name))

- **CourseContent** (id PK, courseFolderId FK, type, url)

7. Normalization

- **1NF:** All attributes contain atomic values
- **2NF:** Partial dependencies are eliminated using composite uniqueness where required
- **3NF:** Non-key attributes depend only on the primary key, with transitive dependencies removed through separate entities

8. Database Implementation

8.1 SQL Implementation:

- DDL commands such as CREATE TABLE, ALTER TABLE, and constraint definitions
- DML operations for inserting, updating, deleting, and retrieving data
- Relational queries involving joins and filtered selections

8.2 PL/SQL Components:

Currently, PL/SQL components such as stored procedures, functions, triggers, cursors, and exception handling are not implemented. These can be incorporated in future iterations to enhance DBMS depth and automation.

9. Transaction Management & Concurrency

Explicit transaction blocks are not manually defined at present. However, data consistency is ensured through atomic operations and enforced database constraints. The DBMS handles concurrency control using locking mechanisms, ensuring consistency.

10. Tools & Technologies Used

- **Database:** PostgreSQL (Prisma ORM)
- **Query Language:** SQL
- **Backend Framework:** Node.js with Express.js
- **Frontend Fra:** React.js
- **Integrations:** AWS S3, Razorpay

11. Expected Outcomes:

The expected outcome of this project is a fully normalized, relational database-backed SaaS course platform that supports secure role-based workflows, efficient data retrieval, structured content management, and reliable enrollment handling. The system ensures improved data consistency, integrity, and scalability.